

Non-Functional Requirements (NFR) — Sprint 3

1. Top 3 Prioritized Non-Functional Requirements

For Sprint 3, our team prioritized the following three non-functional requirements:

1.1 Security

Security was prioritized because the system handles sensitive user information, including patient profiles, appointment details, health metrics, and simulated payment-related data. Since the platform serves different roles such as patients, therapists, and administrators, protecting private data and restricting unauthorized access are essential.

To address security, the team implemented authentication for protected routes, role-based access control, and input validation. These measures help ensure that users can only access the data and actions appropriate to their roles.

1.2 Performance

Performance was prioritized because the system includes booking workflows and assistive device purchasing workflows that must remain responsive for users. Slow response times during booking or payment confirmation would reduce the usability and reliability of the platform.

To address performance, the team optimized key backend queries, reduced unnecessary repeated requests, and ensured that booking conflict checks and purchase-related operations could be completed efficiently. This helps the platform maintain acceptable responsiveness during normal usage.

1.3 Usability

Usability was prioritized because the platform is intended for multiple user groups, including patients who may not be highly technical. The team wanted to ensure that common workflows such as browsing therapists, booking appointments, purchasing assistive devices, and viewing confirmation information are simple and clear.

To improve usability, the team designed straightforward navigation flows, clear confirmation and error messages, and consistent page layouts across core workflows. This reduces user confusion and improves overall accessibility of the platform.

2. Trade-offs Made to Prioritize These NFRs

2.1 Trade-off: Usability vs Advanced Functionality

To prioritize usability, the team simplified workflows instead of implementing more advanced features such as real payment gateway integration and advanced analytics dashboards.

The team is okay with this trade-off because a simple and intuitive interface is more important for users and for the sprint demo.

The benefit we received in return is improved usability, easier navigation, and fewer user errors during booking and purchasing workflows.

2.2 Trade-off: Performance vs Maintainability

To prioritize performance, the team reduced some detailed logging and avoided adding extra processing steps in the booking and purchasing workflows.

This is a trade-off between performance and maintainability, since detailed logging and extra checks would make the system easier to debug and maintain, but would also slow down system response time.

The team is okay with this trade-off because system responsiveness and smooth user experience are more important for the current release.

The benefit we received in return is faster response times and smoother system performance during booking and purchasing operations.